

Skill -03

Part -01:

Code:

```
GNU nano 7.2
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <termios.h>

#define MAX_HISTORY 10
#define MAX_COMMAND 100

void disableRawMode(struct termios *old)
{
    tcsetattr(STDIN_FILENO, TCSAFLUSH, old);
}

void enableRawMode(struct termios *old)
{
    struct termios raw;

    tcgetattr(STDIN_FILENO, old);
    raw = *old;

    raw.c_lflag &= ~(ECHO | ICANON);

    tcsetattr(STDIN_FILENO, TCSAFLUSH, &raw);
}

int main()
{
    char history[MAX_HISTORY][MAX_COMMAND];
    char command[MAX_COMMAND];
    int history_count = 0;
    int history_pos = 0;
    int len = 0;
    char c;

    struct termios old;

    enableRawMode(&old);

    while (1)
    {
        printf("> ");
        fflush(stdout);

        len = 0;
        history_pos = history_count;
    }
}
```

GNU nano 7.2

```
    history_pos = history_count;

    while (1)
    {
        read(STDIN_FILENO, &c, 1);

        if (c == '\n' || c == '\r')
        {
            command[len] = '\0';
            printf("\n");

            break;
        }

        if (c == 127)
        {
            if (len > 0)
            {
                len--;
                printf("\b \b");
                fflush(stdout);
            }
            continue;
        }

        if (c == 27)
        {
            char seq[2];

            read(STDIN_FILENO, &seq[0], 1);
            read(STDIN_FILENO, &seq[1], 1);

            if (seq[0] == '[' && seq[1] == 'A')
            {
                if (history_pos > 0)
                {
                    history_pos--;

                    printf("\33[2K\r> ");
                    strcpy(command, history[history_pos]);
                    len = strlen(command);

                    printf("%s", command);
                    fflush(stdout);
                }
            }
        }
    }
}
```

```
bharath sai reddy@LAPTOP-1  X + v
Default: bharath_sai_reddy@LAPTOP-1711DF88:
~/90128_OSSP/2520090128_Skill/Skill-03
ctrl+alt+1

        len = 0;
        command[0] = '\0';
    }

    fflush(stdout);
}

    continue;
}

    if (len < MAX_COMMAND - 1)
    {
        command[len++] = c;
        putchar(c);
        fflush(stdout);
    }
}

    if (strcmp(command, "exit") == 0)
        break;

    if (len > 0)
    {
        if (history_count < MAX_HISTORY)
        {
            strcpy(history[history_count], command);
            history_count++;
        }
        else
        {
            for (int i = 1; i < MAX_HISTORY; i++)
                strcpy(history[i - 1], history[i]);

            strcpy(history[MAX_HISTORY - 1], command);
        }
    }

    disableRawMode(&old);

    return 0;
}
```

Output:

```
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Skill/Skill-03$ gcc history.c -o history
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Skill/Skill-03$ ./history
> ls
> pwd
> date
> whoami
> exit

Command History:
ls
pwd
date
whoami
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Skill/Skill-03$ nano history.c
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Skill/Skill-03$ gcc history.c -o history
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Skill/Skill-03$ ./history
> ls
> pwd
> date
> date
> exit
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Skill/Skill-03$ |
```

Part -02:

Code:



bharath_sai_reddy@LAPTOP-1



GNU nano 7.2

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

#define COMMAND_SIZE 100

int main()
{
    char **history;
    int count = 0;
    int capacity = 2;

    history = malloc(capacity * sizeof(char *));

    if (history == NULL)
    {
        printf("Memory allocation failed\n");
        return 1;
    }

    while (1)
    {
        char buffer[COMMAND_SIZE];

        printf("> ");
        fgets(buffer, COMMAND_SIZE, stdin);

        buffer[strcspn(buffer, "\n")] = '\0';

        if (strcmp(buffer, "exit") == 0)
            break;

        if (count == capacity)
        {
            capacity *= 2;

            history = realloc(history,
                              capacity * sizeof(char *));

            if (history == NULL)
            {
                printf("Reallocation failed\n");
                return 1;
            }
        }

        history[count] = malloc(
            (strlen(buffer) + 1) * sizeof(char)
        );

        if (history[count] == NULL)
        {
            printf("Memory allocation failed\n");
            return 1;
        }

        strcpy(history[count], buffer);

        count++;
    }
}
```

```

        capacity *= 2;

        history = realloc(history,
                           capacity * sizeof(char *));

        if (history == NULL)
        {
            printf("Reallocation failed\n");
            return 1;
        }

        history[count] = malloc(
            (strlen(buffer) + 1) * sizeof(char)
        );

        if (history[count] == NULL)
        {
            printf("Memory allocation failed\n");
            return 1;
        }

        strcpy(history[count], buffer);

        count++;
    }

    printf("\nCommand History:\n");

    for (int i = 0; i < count; i++)
    {
        printf("%d: %s\n", i + 1, history[i]);
    }

    for (int i = 0; i < count; i++)
        free(history[i]);

    free(history);

    return 0;
}
|

```

 ^G Help	 ^O Write Out	 ^W Where Is	 ^K Cut
 ^X Exit	 ^R Read File	 ^\ Replace	 ^U Paste

Output:

```

bharath_sai_reddy@LAPTOP-1  ×  +  ▾
Preparing to unpack .../4-libsource-highlight-common_3.1.9-4.3build1_all.deb ...
Unpacking libsource-highlight-common (3.1.9-4.3build1) ...
Selecting previously unselected package libsource-highlight4t64:amd64.
Preparing to unpack .../5-libsource-highlight4t64_3.1.9-4.3build1_amd64.deb ...
Unpacking libsource-highlight4t64:amd64 (3.1.9-4.3build1) ...
Selecting previously unselected package gdb.
Preparing to unpack .../6-gdb_15.1-1ubuntu1~24.04.1_amd64.deb ...
Unpacking gdb (15.1-1ubuntu1~24.04.1) ...
Selecting previously unselected package libc6-dbg:amd64.
Preparing to unpack .../7-libc6-dbg_2.39-0ubuntu8.9_amd64.deb ...
Unpacking libc6-dbg:amd64 (2.39-0ubuntu8.9) ...
Selecting previously unselected package valgrind.
Preparing to unpack .../8-valgrind_1%3a3.22.0-0ubuntu3_amd64.deb ...
Unpacking valgrind (1:3.22.0-0ubuntu3) ...
Setting up libdebuginfod-common (0.190-1.1ubuntu0.1) ...
Setting up libsource-highlight-common (3.1.9-4.3build1) ...
Setting up libc6-dbg:amd64 (2.39-0ubuntu8.9) ...
Setting up libipt2 (2.0.6-1build1) ...
Setting up libbabeltrace1:amd64 (1.5.11-3build3) ...
Setting up valgrind (1:3.22.0-0ubuntu3) ...
Setting up libdebuginfod1t64:amd64 (0.190-1.1ubuntu0.1) ...
Setting up libsource-highlight4t64:amd64 (3.1.9-4.3build1) ...
Setting up gdb (15.1-1ubuntu1~24.04.1) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.9) ...
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Skill/Skill-03$ valgrind --leak-check=full ./history_dynamic
==5259== Memcheck, a memory error detector
==5259== Copyright (C) 2002-2022, and GNU GPL'd, by Julian Seward et al.
==5259== Using Valgrind-3.22.0 and LibVEX; rerun with -h for copyright info
==5259== Command: ./history_dynamic
==5259==
> ls
> pwd
> date
> exit

Command History:
1: ls
2: pwd
3: date
==5259==
==5259== HEAP SUMMARY:
==5259==    in use at exit: 0 bytes in 0 blocks
==5259==   total heap usage: 7 allocs, 7 frees, 2,108 bytes allocated
==5259==
==5259== All heap blocks were freed -- no leaks are possible
==5259==
==5259== For lists of detected and suppressed errors, rerun with: -s
==5259== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
bharath_sai_reddy@LAPTOP-1711DF88:~/90128_OSSP/2520090128_Skill/Skill-03$ |

```